

Question No:6

DS

Session

Sorting

Question Information

Level 1 Challenge 16

Question description

Selvan studies, engineering as per his father's wishes, while Aaron, whose family is poor, studies engineering to improve his family's financial situation. sumanth, however, studies engineering of his simple passion for developing data structure applications.

sumanth is participating in a hackathon for data structure application development.

sumanth task is to use Insertion Sort to sort the supplied set of numbers.

As a result, The input provides the number of components on the first line and the numbers to be sorted on the second line. Print the array's state at the third iteration and the final sorted array in the supplied format in the output.

Judge will determine whether the outcome is correct or not.

Can you help him ?

Constraints

$1 \leq N \leq 10^5$

$1 \leq A_i \leq 10^9$

Input Format:

The first line of the input contains the number of elements

the second line of the input contains the numbers to be sorted.

Output Format:

Fist line indicates print the status of the array at the 3rd iteration

second line print the the final sorted array in the given format.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5
64 25 22 90 35

EXPECTED OUTPUT

22 25 64 90 35
22 25 35 64 90

Test Case 2

INPUT (STDIN)

5
16 21 22 10 35

EXPECTED OUTPUT

16 21 22 10 35
10 16 21 22 35

▼ Mandatory Test Cases

Test Case 1

KEYWORD

void insertionSort(int
arr[],int n)

Test Case 2

KEYWORD

void printArray(int arr[],int
n)

Test Case 3

KEYWORD

insertionSort(arr, n);

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

6

Test Case 2

TOKEN COUNT

275

Test Case 3

NLOC

50

Code:

```
1  #include <stdio.h>
2  int main()
3  {
4      int n;
5      scanf("%d", &n);
6      int a[n];
7      for (int i = 0; i < n; i++)
8      {
9          scanf("%d", &a[i]);
10     }
11
12     for (int i = 1; i < n; i++)
13     {
14         /* Print array at the start of 3rd iteration */
15         if (i == 3)
16         {
17             for (int k = 0; k < n; k++)
18             {
19                 printf("%d", a[k]);
20
21                 if (k < n - 1)
22                     printf(" ");
23             }
24             printf("\n");
25         }
26
27         int key = a[i];
28         int j = i - 1;
```

```

29
30     while (j >= 0 && a[j] > key)
31     {
32         a[j + 1] = a[j];
33         j--;
34     }
35
36     a[j + 1] = key;
37 }
38 /* Final sorted array */
39 for (int i = 0; i < n; i++)
40 {
41     printf("%d", a[i]);
42
43     if (i < n - 1)
44         printf(" ");
45 }
46 printf("\n");
47 return 0;
48 }

```

Output:

Test Case-1:

```

5
64 25 22 90 35
22 25 64 90 35
22 25 35 64 90

```

Test Case-2:

```
5
16 21 22 10 35
16 21 22 10 35
10 16 21 22 35
PS C:\Users\kesha\OneDrive\Documents\3sem\C>
```

Keshav Kumar Rauniyar
CH.SC.U4CSE25236